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GASTROINTESTINAL MOTILITY

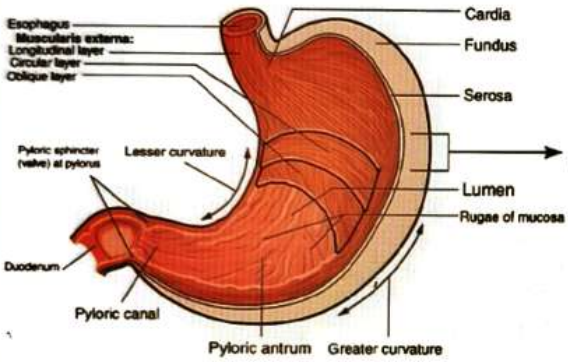
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Interstitial cells of Cajal

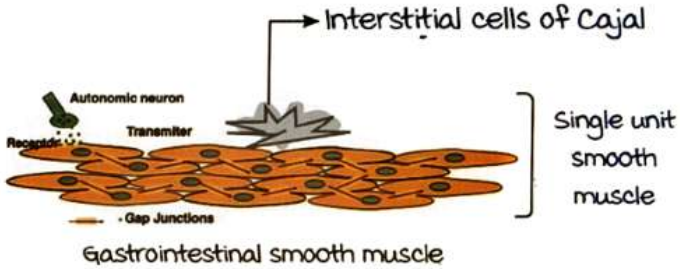
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- Pacemaker of GIT.
- Has a 'restless' membrane potential : BER.

Location :



mid body region along greater. curvature

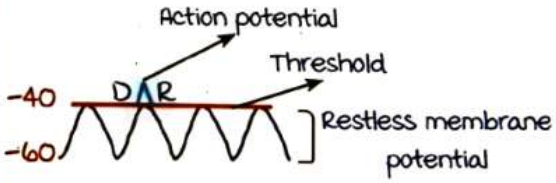


Gastrointestinal smooth muscle

Applied aspect : Gastrointestinal stromal tumor (GIST), arises from interstitial cells of cajal.

Basal electrical rhythm (BER) :

BER	Action Potential/spikes
• Basal potential oscillating b/w −60 to −40. • No role in muscle contraction.	• BER on crossing threshold (−40). • Phases : – Depolarization (D) d/t Ca^{2+} influx. – Repolarization (R) d/t K^+ efflux. • Cause smooth muscle contraction



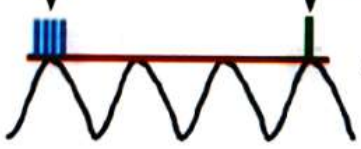
Effect of Autonomic Nervous System (ANS) :

Parasympathetic nervous system (PNS)

Acetylcholine

Sympathetic nervous system (SNS)

Norepinephrine



- Increase in spikes.
  - ↑ force of contraction.
  - ↑ tension generated.
- Decrease in spikes.
  - ↓ force of contraction.
  - ↓ tension generated.

Frequency of BER (Oscillations/min)	
Duodenum	12
Ileum	8
Stomach	4
Caecum	2



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Motility in GIT During Fed State

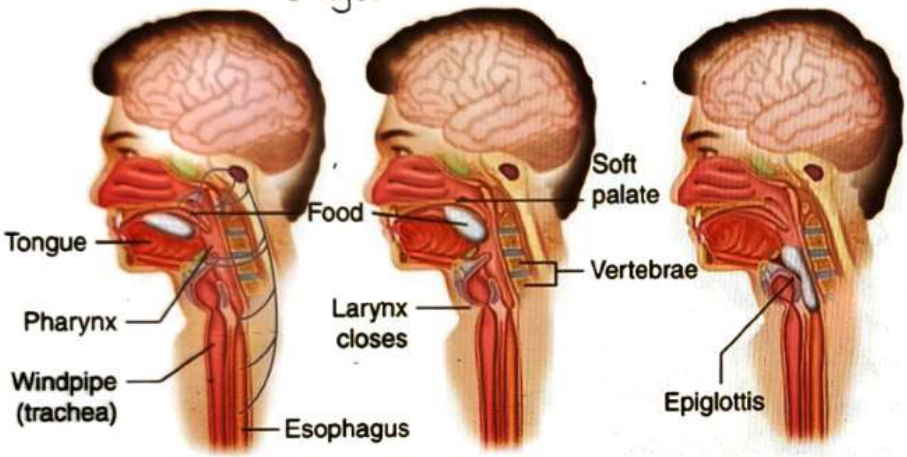
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FOOD IN MOUTH :

Deglutition :

Process of swallowing.

Stages :



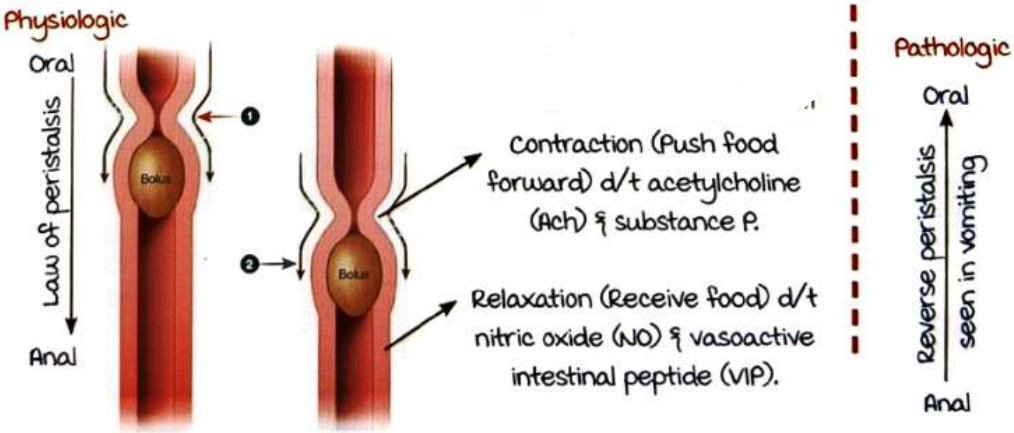
Oral stage (Voluntary) → Pharyngeal Stage (Involuntary) → Esophageal stage (Involuntary)

Deglutition Reflex :
Coordinates swallowing
Food in mouth (Stimulus)
↓
CN V, IX, X (Afferent)
↓
Nucleus tractus solitarius (NTS) in medulla (Centre)
↓
CN V, VII, XII (Efferent)
↓
Innervates pharyngeal muscle & tongue (Effector)
↓
Swallowing (Response)

FOOD IN ESOPHAGUS :

Peristalsis :

Mechanism : Esophageal stretch → Release serotonin → Activates myenteric plexus → Starts peristalsis



Types :

Types	movement	Seen in
Primary	Progressive	Physiologic
Secondary	Progressive	Esophageal irritation
Tertiary	Non Progressive	Esophageal Spasm

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Clinical Aspect :

Absence of nitric oxide

↓

Ach overactivity

↓

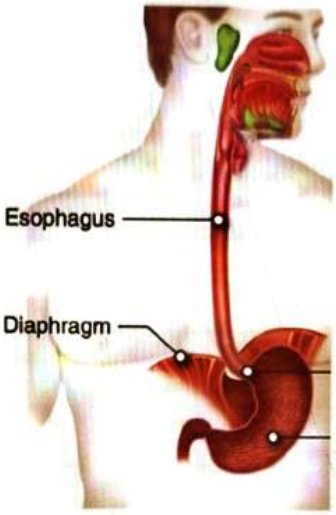
Sustained contraction of LES

↓

Achalasia cardia

Rx : Botulinum toxin (BoTox)

- Inhibit Ach release.



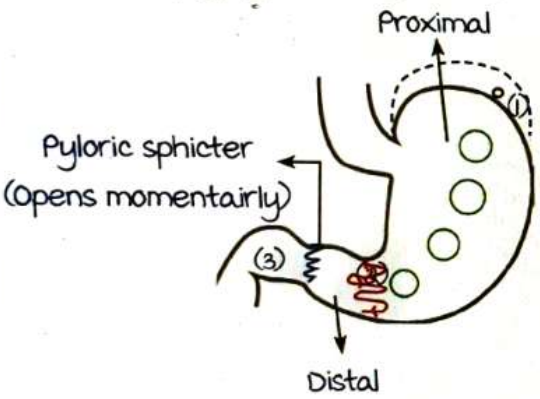
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Normal physiology : [Ach] = [NO]

Lower esophageal sphincter (LES)

FOOD IN STOMACH :

Gastric emptying :



1. Receptive relaxation
   Use : For temporary storage of food.
2. Retropulsion
   Use : Mixing & grinding.
3. Gastric emptying (GE).

Factors affecting GE :

Factors ↑	Factors ↓
• ↑ gastric volume. • Gastrin. • motilin. • Acetylcholine. • Substance P & K. • Serotonin.	• ↑ carbohydrate, fat, protein, acid, hyperosmolar content in duodenum ↓ ⊕ Enteric gastro reflex → Inhibit gastric emptying (Local enteric reflex at GIT). • Nitric oxide. • VIP. • Cholecystokinin. • Dopamine. • Enkephalin.

Clinical aspect :

Gastrocolic reflex :

- Defecation in newborn, following a meal.
- Reflex : Stomach distension (D/t food) → ↑ colonic motility → Defecation.
- Amplified by : Gastrin.

Feedback

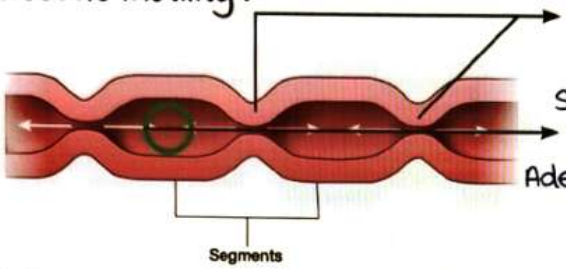
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FOOD IN INTESTINE :

Small intestine motility :



Both ends contract
↓
Slow movement of food
↓
Adequate time for digestion & absorption

Segmentation :

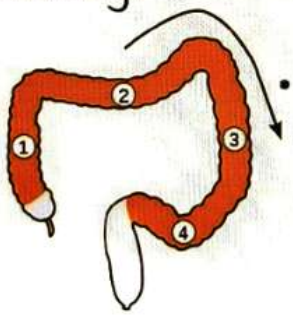
Factors affecting motility.

Factors ↑ motility	Factors ↓ motility
PNS	SNS
Motilin	Secretin
Gastrin	
Cholecystokinin	
Serotonin	
Gastroileal reflex	

Large intestine motility :

Haustrations :

- Combined contraction of circular & longitudinal muscles.
- use : Absorption of water & electrolytes.

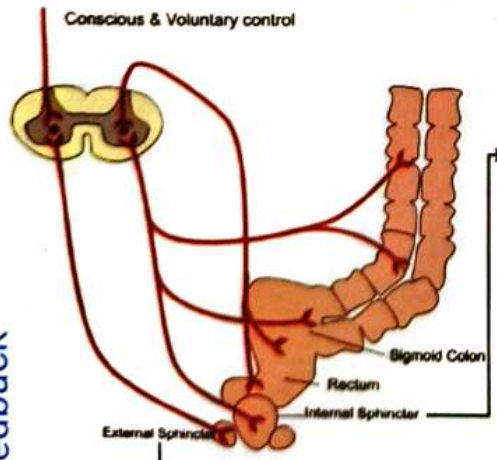


- Mass movement : moves along segment
  - ↓
  - rectal distension
  - ↓
  - ↑ motility
  - ↓
  - Defecation

(1) Proximal colon
(2) (3) (4) : Distal colon

Rectal motility :

Defecation Reflex :



Muscle : Smooth muscle
Nature : Involuntary
Effect of ANS :

- PNS : Relaxes (Facilitates defecation).
- SNS : Contracts (Inhibits defecation).

Muscle : Skeletal muscle.
Nature : Voluntary.
Nerve : Pudendal nerve.

Mechanism :

Rectal distension
↓ Via stretch afferent
Spinal cord
↓ PNS output
Relaxes Internal anal sphincter (IAS)
↓
If situation favourable :

- Powerful contractions of abdominal muscles, rectum & colon.
- Relaxation of external anal sphincter (EAS).

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Rectal Pressure :

18 mmHg : 1st urge to defecate.

55 mmHg : Voluntary control lost.

Sudden, inadvertent release of IAS & EAS

↓

Soiling

Note :

Transit Time	
Esophagus	2-3 seconds
Stomach	2-5 hours
Small Intestine	3-6 hours
Caecum	4 hrs
Proximal 1/3rd of colon	6 hrs
Distal 2/3rd of colon	9 hrs
Sigmoid colon	12 hrs

----- Active space -----

Note :

- Dietary fibres/roughage :
  - Eg : Lignin, pectin, cellulose, hemicellulose.
  - Source : Cereal grains, fruits & vegetables.
  - Functions :
    a. Add bulk to diet (↓ food intake).
    b. Slow absorption of glucose.
    c. Reduces total & LDL cholesterol.
    d. Add bulk to stool (Retains water).
    e. Roughage $\xrightarrow{\text{microflora}}$ Short-chain fatty acids.
    f. Prevent colonic diverticulum.

Motility in GIT during Fasting State

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MIGRATORY MOTOR COMPLEX (MMC) :

Seen in : Fasting & interdigestive period.

Mechanism : Stomach $\xrightarrow{\text{MMC}}$ Colon.

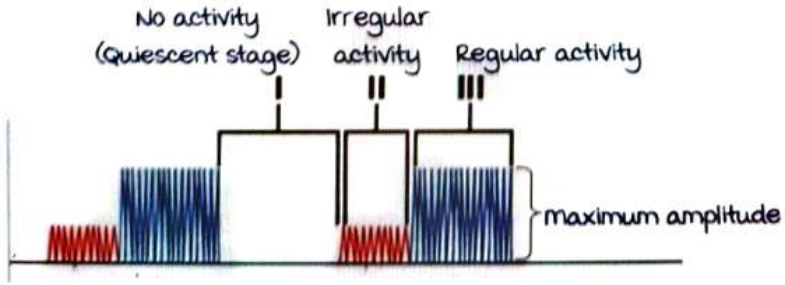
Occurrence : Occurs once in 90 minutes (Rate : 5cm/min).

Regulation : Motilin/time keeper (Released from Mo cells).

- ↑ GI motility.
- Erythromycin : Acts via motilin receptor.
  Used for Rx of gastroparesis in diabetes.

Function : Helps in clearing foreign body, bacteria & undigested food.

Phases :



Feedback